1) TIC TAC TOE

```
board = [' ' for _ in range(9)]

def print_board(board):
    print(f"\n {board[0]} | {board[1]} | {board[2]} ")
    print("-----------")
    print(f" {board[3]} | {board[4]} | {board[5]} ")
    print("-----------")
    print(f" {board[6]} | {board[7]} | {board[8]} \n")

def check_winner(b, player):
    win_conditions = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ]
    return any(all(b[i] == player for i in cond) for cond in win_conditions)

def is_full(b):
    return ' ' not in b

def minimax(b, is_maximizing):
    if check_winner(b, 'O'): return 1
    if check_winner(b, 'X'): return -1
    if is_full(b): return 0

    if is_maximizing:
        best_score = -float('inf')
        for i in range(9):
            if b[i] == ' ' :
                b[i] = 'O'
                score = minimax(b, False)
                b[i] = ' '
                best_score = max(score, best_score)
        return best_score
    else:
        best_score = float('inf')
        for i in range(9):
            if b[i] == ' ':
                b[i] = 'X'
                score = minimax(b, True)
```

```
            b[i] = ' '
            best_score = min(score, best_score)
        return best_score

def ai_move(b):
    best_score = -float('inf')
    move = None
    for i in range(9):
        if b[i] == ' ':
            b[i] = 'O'
            score = minimax(b, False)
            b[i] = ' '
            if score > best_score:
                best_score = score
                move = i
    b[move] = 'O'


print("You are X, AI is O.")
print_board(board)

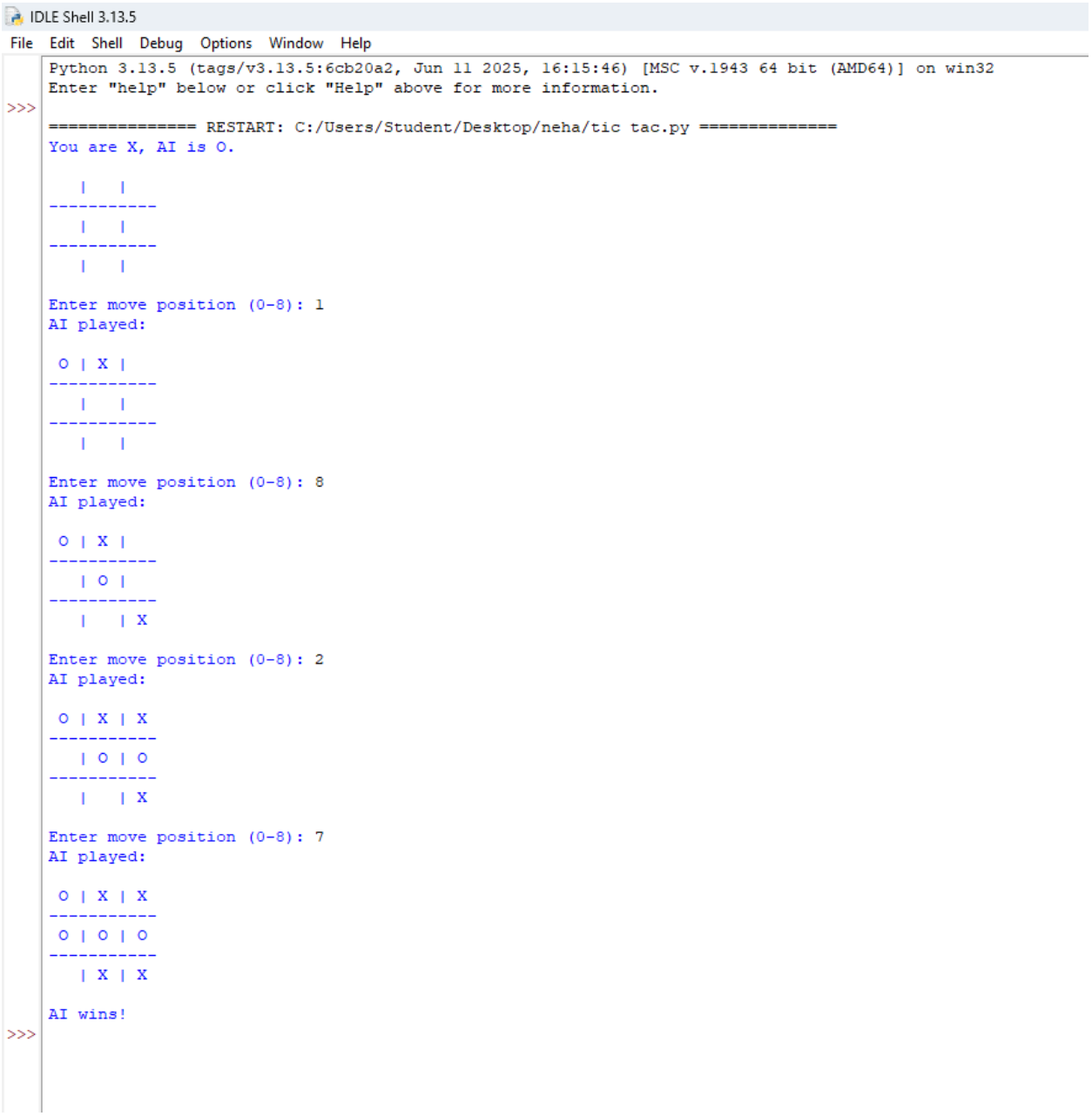
while True:

    pos = int(input("Enter move position (0-8): "))
    if board[pos] == ' ':
        board[pos] = 'X'
    else:
        print("Invalid move! Try again.")
        continue

    if check_winner(board, 'X'):
        print_board(board)
        print("You win!")
        break
    if is_full(board):
        print_board(board)
        print("It's a tie!")
        break


    ai_move(board)
    print("AI played:")
    print_board(board)
```

```
    if check_winner(board, 'O'):
        print("AI wins!")
        break
    if is_full(board):
        print("It's a tie!")
        break
```



2) Vacuum Cleaner

```
def print_status(room_a, room_b, position):
    print("\n--- Vacuum Cleaner ---")
    print("Room A:", room_a)
    print("Room B:", room_b)
    print("Vacuum Position:", position)


def vacuum_cleaner(room_a, room_b, position):

    print_status(room_a, room_b, position)

    if position == 'A':

        if room_a == 'Dirty':
            print("Room A is Dirty -> SUCK")
            room_a = 'Clean'
        else:
            print("Room A is Clean -> IDLE")

        print("Moving from A to B")
        position = 'B'

        if room_b == 'Dirty':
            print("Room B is Dirty -> SUCK")
            room_b = 'Clean'
        else:
            print("Room B is Clean -> IDLE")

    else:

        if room_b == 'Dirty':
            print("Room B is Dirty -> SUCK")
            room_b = 'Clean'
        else:
            print("Room B is Clean -> IDLE")

        print("Moving from B to A")
        position = 'A'

        if room_a == 'Dirty':
            print("Room A is Dirty -> SUCK")
```

```
            room_a = 'Clean'
        else:
            print("Room A is Clean -> IDLE")

    print("\nFinal Status:")
    print_status(room_a, room_b, position)


# Input

room_a = input("Enter status of Room A (Dirty/Clean): ")
room_b = input("Enter status of Room B (Dirty/Clean): ")
position = input("Enter vacuum position (A/B): ")

vacuum_cleaner(room_a, room_b, position)
```

